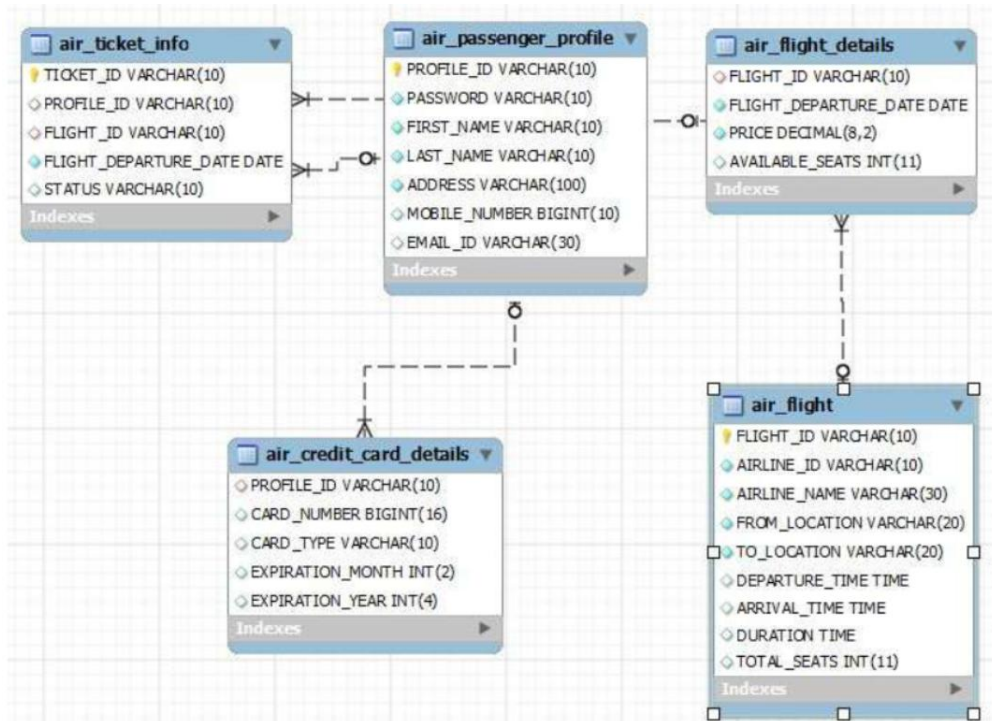


DAY-3 Assignment

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```
CREATE TABLE air_passenger_profile (  
    profile_id VARCHAR(10) PRIMARY KEY,  
    password VARCHAR(10),  
    first_name VARCHAR(10),  
    last_name VARCHAR(10),  
    address VARCHAR(100),  
    mobile_number BIGINT,  
    email_id VARCHAR(30)  
);  
  
CREATE TABLE air_flight (  
    flight_id VARCHAR(10) PRIMARY KEY,  
    airline_id VARCHAR(10),  
    airline_name VARCHAR(30),  
    from_location VARCHAR(20),  
    to_location VARCHAR(20),  
    departure_time TIME,  
    arrival_time TIME,  
    duration TIME,  
    total_seats INT  
);  
  
CREATE TABLE air_flight_details (  
    flight_id VARCHAR(10),  
    flight_departure_date DATE,  
    price DECIMAL(8,2),  
    available_seats INT,  
    PRIMARY KEY (flight_id, flight_departure_date),  
    FOREIGN KEY (flight_id)
```

```

        REFERENCES air_flight(flight_id)
    );
CREATE TABLE air_ticket_info (
    ticket_id VARCHAR(10) PRIMARY KEY,
    profile_id VARCHAR(10),
    flight_id VARCHAR(10),
    flight_departure_date DATE,
    status VARCHAR(10),
    FOREIGN KEY (profile_id)
        REFERENCES air_passenger_profile(profile_id),
    FOREIGN KEY (flight_id, flight_departure_date)
        REFERENCES air_flight_details(flight_id, flight_departure_date)
);
CREATE TABLE air_credit_card_details (
    profile_id VARCHAR(10),
    card_number BIGINT,
    card_type VARCHAR(10),
    expiration_month INT,
    expiration_year INT,
    PRIMARY KEY (profile_id, card_number),
    FOREIGN KEY (profile_id)
        REFERENCES air_passenger_profile(profile_id)
);

```

```

INSERT INTO air_passenger_profile VALUES
('P001','pass1','Anu','Varma','Hyderabad',9876543210,'anu@gmail.com'),
('P002','pass2','Ravi','Kumar','Delhi',9123456780,'ravi@gmail.com'),
('P003','pass3','Sita','Rao','Mumbai',9988776655,'sita@gmail.com'),
('P004','pass4','Aman','Shah','Pune',9090909090,'aman@gmail.com'),
('P005','pass5','Neha','Singh','Chennai',9012345678,'neha@gmail.com');

```

```

INSERT INTO air_flight VALUES
('F001','A001','IndiGo','Hyderabad','Delhi','06:00','08:00','02:00',180),
('F002','A002','AirIndia','Mumbai','Bangalore','09:00','11:00','02:00',160),
('F003','A003','Vistara','Chennai','Kolkata','14:00','17:00','03:00',150),
('F004','A004','SpiceJet','Pune','Hyderabad','18:00','19:30','01:30',140),
('F005','A005','Akasa','Delhi','Mumbai','20:00','22:00','02:00',170);

```

```

INSERT INTO air_flight VALUES
('F009','A009','IndiGo','Chennai','Hyderabad','06:45','08:15','01:30',160);

```

```

INSERT INTO air_flight_details VALUES
('F009','2025-04-25',4300.00,120);

```

```

INSERT INTO air_ticket_info VALUES
('T009','P001','F009','2025-04-25','Booked');

```

```

INSERT INTO air_flight_details VALUES
('F001','2025-01-05',4500.00,120),
('F002','2025-01-06',5200.00,100),

```

```
('F003','2025-01-07',6000.00,90),
('F004','2025-01-08',3500.00,110),
('F005','2025-01-09',4800.00,130);
```

```
INSERT INTO air_ticket_info VALUES
('T001','P001','F001','2025-01-05','Booked'),
('T002','P002','F002','2025-01-06','Booked'),
('T003','P003','F003','2025-01-07','Cancelled'),
('T004','P004','F004','2025-01-08','Booked'),
('T005','P005','F005','2025-01-09','Booked');
```

```
INSERT INTO air_credit_card_details VALUES
('P001',1111222233334444,'VISA',12,2026),
('P002',2222333344445555,'MASTER',11,2027),
('P003',3333444455556666,'VISA',10,2028),
('P004',4444555566667777,'RUPAY',9,2026),
('P005',5555666677778888,'VISA',8,2029);
select * from air_flight_details;
```

```
INSERT INTO air_ticket_info VALUES
('T006','P001','F002','2025-01-06','Booked'),
('T007','P001','F003','2025-01-07','Booked'),
('T008','P002','F004','2025-01-08','Booked');
```

```
INSERT INTO air_flight VALUES
('F006','A006','IndiGo','Hyderabad','Mumbai','07:30','09:30','02:00',180)
,
('F007','A007','AirIndia','Delhi','Chennai','12:00','15:00','03:00',170),
('F008','A008','Vistara','Bangalore','Kolkata','18:30','21:30','03:00',160);
```

```
INSERT INTO air_flight_details VALUES
('F006','2025-02-12',4900.00,140),
('F007','2025-03-18',5600.00,130),
('F008','2025-04-22',6100.00,125);
```

```
1.
select f.flight_id,
f.from_location,f.to_location,datetime(month,d.flight_departure_date) as
Month_Name,avg(d.price) as Average_Price
from air_flight f
join air_flight_details d on d.flight_id=f.flight_id
group by
datetime(month,d.flight_departure_date),f.flight_id,f.from_location,f.to_
location,month(d.flight_departure_date)
order by f.flight_id,month(d.flight_departure_date);
```

Results		Messages			
	flight_id	from_location	to_location	Month_Name	Average_Price
1	F001	Hyderabad	Delhi	January	4500.000000
2	F002	Mumbai	Bangalore	January	5200.000000
3	F003	Chennai	Kolkata	January	6000.000000
4	F004	Pune	Hyderabad	January	3500.000000
5	F005	Delhi	Mumbai	January	4800.000000
6	F006	Hyderabad	Mumbai	February	4900.000000
7	F007	Delhi	Chennai	March	5600.000000
8	F008	Bangalore	Kolkata	April	6100.000000
9	F009	Chennai	Hyderabad	April	4300.000000

2.

```
select p.profile_id,p.first_name,p.last_name,p.address,
count(i.ticket_id) as No_of_Tickets from air_passenger_profile p
join air_ticket_info i on i.profile_id=p.profile_id
group by p.profile_id,p.first_name,p.last_name,p.address
having count(i.ticket_id)=(select top 1 count(ticket_id) from
air_ticket_info group by profile_id
order by count(ticket_id))
order by p.first_name;
```

	profile_id	first_name	last_name	address	No_of_Tickets
1	P004	Aman	Shah	Pune	1
2	P005	Neha	Singh	Chennai	1
3	P003	Sita	Rao	Mumbai	1

3.

```
select
f.from_location,f.to_location,datename(month,d.flight_departure_date) as
Month_Name,count(d.flight_departure_date) as No_of_services from
air_flight f
join air_flight_details d on d.flight_id=f.flight_id
group by
datename(month,d.flight_departure_date),f.from_location,f.to_location
order by
f.from_location,f.to_location,datename(month,d.flight_departure_date);
```

	from_location	to_location	Month_Name	No_of_services
1	Bangalore	Kolkata	April	1
2	Chennai	Hyderabad	April	1
3	Chennai	Kolkata	January	1
4	Delhi	Chennai	March	1
5	Delhi	Mumbai	January	1
6	Hyderabad	Delhi	January	1
7	Hyderabad	Mumbai	February	1
8	Mumbai	Bangalore	January	1
9	Pune	Hyderabad	January	1

4.

```
select p.profile_id,p.first_name,p.last_name,p.address,
count(i.ticket_id) as No_of_Tickets from air_passenger_profile p
join air_ticket_info i on i.profile_id=p.profile_id
group by p.profile_id,p.first_name,p.last_name,p.address
having count(i.ticket_id)=(select top 1 count(ticket_id) from
air_ticket_info group by profile_id
order by count(ticket_id) desc)
order by p.first_name;
```

	profile_id	first_name	last_name	address	No_of_Tickets
1	P001	Anu	Varma	Hyderabad	4

5.

```
select
p.profile_id,p.first_name,p.last_name,i.flight_id,i.flight_departure_date
, count(i.ticket_id) as No_of_Tickets from air_passenger_profile p
join air_ticket_info i on i.profile_id=p.profile_id
join air_flight f on i.flight_id=f.flight_id
where f.from_location='Chennai' and f.to_location='Hyderabad'
group by
p.profile_id,p.first_name,p.last_name,i.flight_id,i.flight_departure_date
order by p.profile_id,i.flight_id;
```

	profile_id	first_name	last_name	flight_id	flight_departure_date	No_of_Tickets
1	P001	Anu	Varma	F009	2025-04-25	1

6.

```
select f.flight_id,f.from_location,f.to_location,d.price from air_flight
f
join air_flight_details d on d.flight_id=f.flight_id
where datename(month,d.flight_departure_date)='April';
```

	flight_id	from_location	to_location	price
1	F008	Bangalore	Kolkata	6100.00
2	F009	Chennai	Hyderabad	4300.00

7.

```
select f.flight_id, f.from_location,f.to_location,avg(d.price) as
Average_Price
from air_flight f
join air_flight_details d on d.flight_id=f.flight_id
group by f.flight_id, f.from_location,f.to_location
order by f.flight_id,f.from_location,f.to_location;
```

	flight_id	from_location	to_location	Average_Price
1	F001	Hyderabad	Delhi	4500.000000
2	F002	Mumbai	Bangalore	5200.000000
3	F003	Chennai	Kolkata	6000.000000
4	F004	Pune	Hyderabad	3500.000000
5	F005	Delhi	Mumbai	4800.000000
6	F006	Hyderabad	Mumbai	4900.000000
7	F007	Delhi	Chennai	5600.000000
8	F008	Bangalore	Kolkata	6100.000000
9	F009	Chennai	Hyderabad	4300.000000

8.

```
select p.profile_id,p.first_name+', '+p.last_name as
customer_name,p.address from air_passenger_profile p
join air_ticket_info i on i.profile_id=p.profile_id
join air_flight f on i.flight_id=f.flight_id
where f.from_location='Chennai' and f.to_location='Hyderabad'
group by p.profile_id,p.first_name+', '+p.last_name,p.address
order by p.profile_id;
```

	profile_id	customer_name	address
1	P001	Anu,Varma	Hyderabad

9.

```
select profile_id from air_ticket_info
group by profile_id
having count(ticket_id)=(select top 1 count(ticket_id) from
air_ticket_info group by profile_id order by count(ticket_id) desc)
order by profile_id;
```

	Results	Messages
	profile_id	
1	P001	

10.

```
select f.flight_id,f.from_location,f.to_location,count(i.ticket_id) as
No_of_Tickets from air_flight f
join air_ticket_info i on i.flight_id=f.flight_id
group by f.flight_id,f.from_location,f.to_location
order by f.flight_id;
```

Results

Messages

	flight_id	from_location	to_location	No_of_Tickets
1	F001	Hyderabad	Delhi	1
2	F002	Mumbai	Bangalore	2
3	F003	Chennai	Kolkata	2
4	F004	Pune	Hyderabad	2
5	F005	Delhi	Mumbai	1
6	F009	Chennai	Hyderabad	1